
AI-Driven Pan-Cancer Drug Prioritization for ecDNA-amplified Oncogenes through CIViC Evidence Integration

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Abstract

Extrachromosomal DNA (ecDNA) is a major driver of high-copy oncogene amplification, promoting intratumoral heterogeneity and rapid therapeutic adaptation. However, ecDNA is rarely incorporated into precision oncology workflows because its detection requires whole-genome sequencing (WGS) with specialized reconstruction, while most clinical evidence resources are curated at the gene or variant level rather than amplification architecture. We present an evidence-aware AI co-scientist workflow that links ecDNA-amplified oncogenes from pan-cancer WGS data to curated therapeutic evidence in the Clinical Interpretation of Variants in Cancer (CIViC) knowledgebase. Using the Pan-Cancer Analysis of Whole Genomes (PCAWG) dataset, we summarize ecDNA-positive tumor prevalence and recurrent ecDNA-amplified oncogenes across cancer types and systematically map gene-tumor contexts to CIViC therapeutic evidence via structured API queries. Based on evidence availability and match quality, ecDNA-amplified contexts are stratified into context-matched, partially matched, and evidence-sparse categories. The framework generates ranked, auditable candidate-therapy reports and explicitly highlights evidence gaps, providing a reproducible pre-screening tool for translational hypothesis generation and integration of ecDNA biology into precision oncology research.

1 ecDNA in cancer and why it matters for therapy

1.1 ecDNA biology and oncogene dosage

ecDNA refers to circular, extrachromosomal DNA elements that often harbor oncogenes and regulatory sequences, supporting high-level focal amplification and heterogeneous copy-number inheritance. ecDNA can drive rapid tumor evolution and genetic heterogeneity by enabling uneven segregation and dynamic copy number changes across cell divisions.¹⁻³

1.2 Clinical relevance: heterogeneity and drug resistance

ecDNA-based amplification has been associated with aggressive disease biology and poor clinical outcome across multiple cancer types. Mechanistically, ecDNA can accelerate adaptation under therapeutic pressure by amplifying or reshaping oncogene dosage, potentially contributing to acquired resistance and recurrence.³⁻⁵

1.3 Why ecDNA is not yet a routine precision oncology variable

In current clinical workflows, focal amplification is often summarized as a copy-number call without explicit representation of ecDNA topology⁶. Three practical barriers limit routine ecDNA use in precision medicine: (i) ecDNA detection typically requires WGS and dedicated reconstruction pipelines; (ii) most clinical evidence resources are organized around genes, variants, and diseases rather than amplicon architecture; and (iii) ecDNA-specific response, resistance evidence remains comparatively sparse, complicating guideline-level decision rules.

2 Precision oncology evidence matching and CIViC

2.1 Standard precision oncology workflow

A typical precision oncology pipeline proceeds from molecular profiling to variant calling, annotation, evidence matching, and multidisciplinary interpretation (e.g., molecular tumor boards). Actionability depends on the strength and context of clinical evidence, including tumor type, biomarker definition, and line of therapy⁷.

2.2 CIViC as a curated, transparent evidence resource

CIViC (Clinical Interpretation of Variants in Cancer) is an expert-curated, open knowledgebase that links cancer variants to evidence statements across diagnostic, prognostic, predisposing, and predictive (therapeutic) categories. CIViC emphasizes provenance, structured evidence fields, and public APIs, which enable reproducible, auditable evidence retrieval.

2.3 Role of an AI co-scientist in evidence-aware interpretation

An AI co-scientist is most useful when constrained to evidence-aware operations: extracting structured conditions from evidence text, summarizing support versus contradiction, and flagging ambiguous or conflicting items for human review. In this work, AI assistance is explicitly used to (i) normalize evidence descriptions into a consistent memo format and (ii) surface gaps (e.g., tumor-context mismatch, alteration-type mismatch) that limit immediate clinical translation.

3 Methods: PCAWG ecDNA landscape and CIViC-driven drug proposals

Workflow: ecDNA amplified oncogene → CIViC evidence

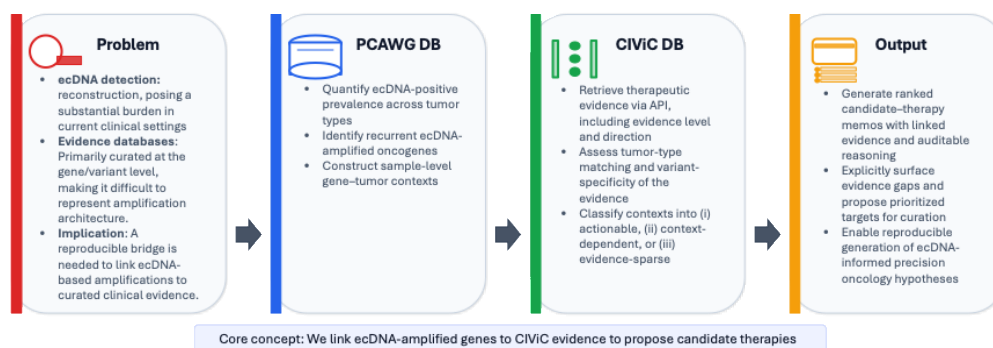


Figure 1. Overview the workflow

3.1 Data sources

We use public PCAWG donor clinical metadata⁸ distributed via an open object-store endpoint, together with established ecDNA/amplicon annotations derived from WGS-based focal amplification reconstruction⁹. CIViC evidence is retrieved programmatically through its public API to ensure reproducibility and traceability (<https://civicdb.org/>).

3.2 ecDNA-amplified oncogene extraction

For ecDNA-positive samples, we compile a sample-by-gene matrix of amplified oncogenes located on ecDNA/amplicons. Where available, amplicon intervals are mapped to gene coordinates to derive gene-level amplification candidates for downstream evidence matching.

3.3 CIViC mapping and evidence normalization

Each candidate gene is queried against CIViC to retrieve therapeutic evidence items and associated metadata (disease context, evidence level, clinical significance, and directionality). Evidence items are normalized into a common representation to support deterministic ranking and review.

3.4 Drug proposal generation: ranked candidate-therapy note

Rather than generating de novo therapeutic recommendations, our framework proposes candidate therapies through structured prioritization of CIViC-supported evidence. For each ecDNA-amplified gene–tumor context, therapeutic evidence retrieved from CIViC is ranked according to explicit and reproducible criteria:

- (i) tumor-context concordance, with evidence from the same or closely related cancer types prioritized;
- (ii) alteration-type relevance, favoring evidence applicable to gene amplification when available;
- (iii) evidence strength, as defined by CIViC evidence level; and
- (iv) evidence directionality, with sensitivity and resistance evidence evaluated separately.

For each sample, the framework generates a standardized output report that summarizes primary candidate therapies, alternative options, contraindication or resistance flags, and explicit evidence gaps that require further expert review or targeted curation. This report format is designed to support transparent inspection of the underlying evidence rather than to provide prescriptive clinical decisions.

We assume that high-copy ecDNA amplification represents a biologically meaningful oncogenic alteration that can be reasonably mapped onto gene-level amplification annotations commonly used in clinical evidence resources. However, this framework does not assert clinical equivalence between ecDNA amplification and other forms of gene amplification. Instead, it provides an evidence-aware and auditable mapping strategy intended for translational hypothesis generation, prioritization, and experimental follow-up.

4 Results

4.1 ecDNA distribution across tumor types

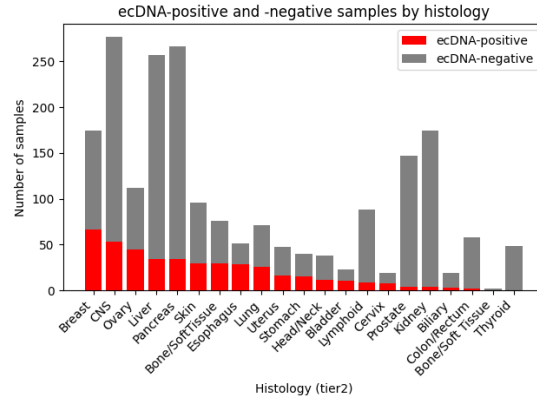


Figure 2. ecDNA sample counts by cancer types in PCAWG

Across pan-cancer cohorts, ecDNA-positive tumors were observed across a wide range of histological cancer types, with substantial variation in prevalence between lineages. Using PCAWG metadata and ecDNA annotations, we quantified the number of ecDNA-positive and ecDNA-negative samples across tier-2 histology categories. ecDNA-positive cases were particularly enriched in tumor types known to experience strong selective pressure for focal oncogene amplification, including ovarian, breast, and central nervous system tumors, while other histologies exhibited lower but non-negligible frequencies (Figure 2). These results highlight the heterogeneous distribution of ecDNA across cancer types and underscore its recurrent association with tumor contexts characterized by aggressive growth and amplified oncogenic signaling.

4.2 Recurrent ecDNA-amplified oncogenes

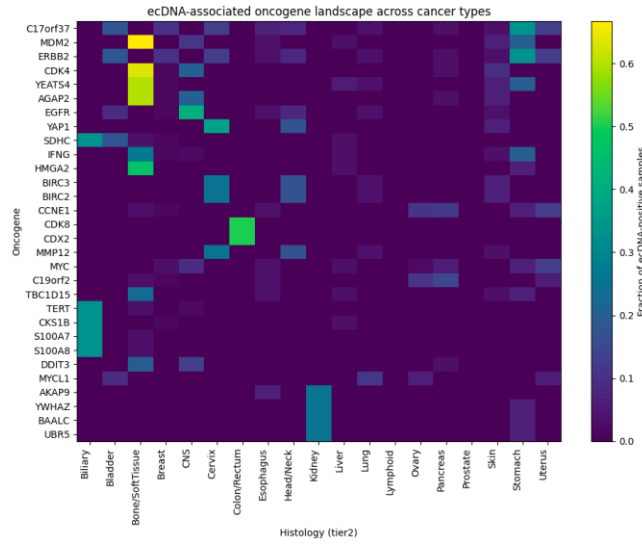


Figure 3. Oncogene landscape from ecDNA-amplified region across tumor types

Prior pan-cancer studies have reported recurrent ecDNA amplification of canonical oncogenic drivers, including EGFR, MYC, MDM2, CDK4, CCND1, and ERBB2, and have demonstrated that ecDNA-mediated amplification can exhibit distinct biological behavior compared to chromosomal amplification. Consistent with these observations, we compiled a pan-cancer landscape of recurrent ecDNA-amplified oncogenes across tier-2 histology categories in PCAWG.

As shown in Figure 3, ecDNA-associated oncogene amplification displays pronounced heterogeneity across tumor types, with certain genes recurrently amplified in specific histological contexts. Canonical drivers such as MDM2 and CDK4 show enrichment in mesenchymal and brain-related tumors, while EGFR and ERBB2 exhibit more lineage-restricted patterns consistent with epithelial tumor contexts. In contrast, other oncogenes demonstrate lower-frequency but recurrent ecDNA amplification across multiple cancer types, highlighting the diverse routes through which ecDNA contributes to oncogenic signaling.

This recurrent ecDNA amplification landscape provides a rational basis for downstream evidence matching, enabling prioritization of gene–tumor contexts for CIViC-based therapeutic interpretation and hypothesis generation.

4.3 CIViC evidence coverage workflow and practical outputs

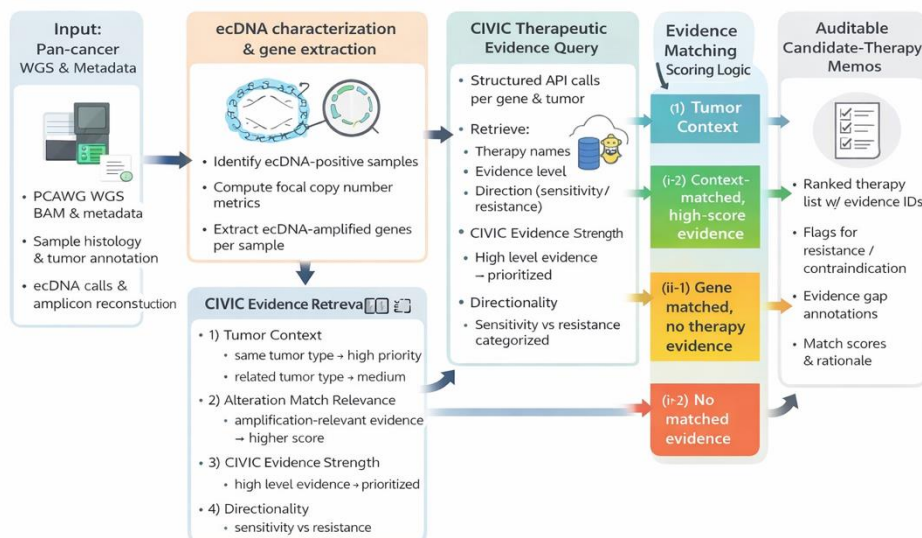


Figure 4. Evidence-aware AI Workflow for ecDNA-Amplified Oncogene Interpretation

(i-1) direct therapeutic evidence in relevant <pre>{'Sample name': 'D028823-SP61453', 'ecDNA_present': True, 'Oncogenes': ['C19orf2', 'CCNE1'], 'ecDNA_copy_number_max': 8.5594104096, 'organ_system': 'OVARY', 'histology_abbreviation': 'Ovary-AdenoCA', 'histology_tier1': 'MESODERM', 'histology_tier2': 'Ovary', 'histology_tier3': 'Adenocarcinoma', 'histology_tier4': 'Serous cystadenocarcinoma', 'CIViC_best_gene': 'CCNE1', 'CIViC_best_disease': 'Ovarian Serous Cystadenocarcinoma', 'CIViC_best_molecular_profile': 'CCNE1 Amplification', 'CIViC_match_score': 7.0, 'Therapies': ['Akt Inhibitor MK2206', 'Dinaciclib']}</pre>	(ii-1) lack of matched therapy information <pre>{'Sample name': 'D051494-SP116949', 'ecDNA_present': True, 'Oncogenes': ['C19orf2', 'CEBPA'], 'ecDNA_copy_number_max': 18.4774838706, 'organ_system': 'PANCREAS', 'histology_abbreviation': 'Panc-AdenoCA', 'histology_tier1': 'ENDODERM', 'histology_tier2': 'Pancreas', 'histology_tier3': 'Adenocarcinoma', 'histology_tier4': 'Adenocarcinoma', 'CIViC_status': 'No matched CIViC evidence'}</pre>
(i-2) direct therapeutic evidence in relevant but low-level match score <pre>{'Sample name': 'D012034-SP25833', 'ecDNA_present': True, 'Oncogenes': ['EGFR'], 'ecDNA_copy_number_max': 158.7183298245, 'organ_system': 'BRAIN, CRANIAL NERVES, & SPINAL CORD', 'histology_abbreviation': 'CNS-GBM', 'histology_tier1': 'NEURAL_CREST', 'histology_tier2': 'CNS', 'histology_tier3': 'Diffuse glioma', 'histology_tier4': 'Glioblastoma', 'CIViC_best_gene': 'EGFR', 'CIViC_best_disease': 'Brain Glioblastoma Multiforme', 'CIViC_best_molecular_profile': 'EGFR Overexpression', 'CIViC_match_score': 3, 'Therapies': ['Afatinib']}</pre>	(ii-2) lack of matched therapy information <pre>{'Sample name': 'D01012-SP2155', 'ecDNA_present': True, 'Oncogenes': ['GAB2', 'INTS4'], 'ecDNA_copy_number_max': 13.3099527576, 'organ_system': 'BREAST', 'histology_abbreviation': 'Breast-AdenoCA', 'histology_tier1': 'ECTODERM', 'histology_tier2': 'Breast', 'histology_tier3': 'Adenocarcinoma', 'histology_tier4': 'Infiltrating duct carcinoma', 'CIViC_status': 'No matched CIViC evidence'}</pre>

Figure 5. Stratification of ecDNA-amplified oncogene-tumor contexts by CIViC therapeutic evidence availability and match quality

Mapping ecDNA-amplified oncogenes to CIViC therapeutic evidence yields a spectrum of practical outcomes that reflect both evidence availability and contextual relevance. We further refine these outcomes into four interpretable categories based on the presence and strength of matched therapeutic evidence.

First, a subset of ecDNA-amplified gene–tumor contexts shows direct therapeutic evidence in relevant disease settings with high contextual concordance, representing cases with clear CIViC-supported therapeutic hypotheses (i-1). A related group exhibits direct therapeutic evidence in relevant tumor contexts but with lower match scores, reflecting partial alignment in disease ontology or biomarker definition (i-2). These cases remain informative but require cautious interpretation and expert review.

In contrast, other ecDNA-amplified contexts lack matched therapeutic information. This includes cases in which no CIViC-supported therapies are available despite successful gene-level matching (ii-1), as well as cases where neither disease- nor alteration-level therapeutic evidence could be matched (ii-2). These evidence-sparse or unmapped contexts highlight systematic gaps in current clinical knowledgebases and represent prioritized targets for curation or targeted literature review.

For each ecDNA-positive sample, the AI co-scientist summarizes the matched evidence into a standardized, auditable memo, explicitly reporting match scores, supporting or contradictory evidence, and unresolved gaps. This structured representation is designed to support expert

inspection, reconciliation, and downstream experimental prioritization rather than automated clinical decision-making.

5 Discussion

ecDNA-aware interpretation offers a complementary perspective to conventional copy-number analysis by explicitly considering amplification architecture associated with intratumoral heterogeneity and rapid evolutionary adaptation. Even when the amplified oncogene is identical, ecDNA topology can imply distinct biological behavior, including dynamic copy-number changes and uneven inheritance, which may motivate alternative monitoring strategies or combination treatment hypotheses in settings where adaptive resistance is expected.

In this study, we demonstrate that ecDNA-amplified oncogenes can be systematically connected to curated precision oncology evidence by leveraging existing gene- and variant-level resources. By mapping ecDNA-positive gene-tumor contexts to CIViC therapeutic annotations and stratifying them according to evidence availability and match quality, our framework operationalizes ecDNA biology within the current evidence ecosystem. Importantly, this approach does not assume clinical equivalence between ecDNA amplification and other forms of gene amplification; instead, it provides an evidence-aware bridge that makes explicit where existing therapeutic knowledge is supportive, ambiguous, or absent.

Several limitations should be acknowledged. Our analysis depends on the availability and quality of ecDNA annotations in public whole-genome sequencing cohorts, which may vary across tumor types and reconstruction pipelines. In addition, therapeutic evidence in CIViC remains unevenly distributed across genes, diseases, and alteration types, with limited coverage of amplification-specific mechanisms and emerging targets. Differences in biomarker definitions and disease ontologies across studies can further constrain automated evidence matching, necessitating careful expert reconciliation in context-dependent cases.

In practice, this framework is intended for use by translational researchers or molecular tumor boards as a pre-screening and prioritization tool, rather than as a clinical decision-making system. By generating standardized, auditable candidate-therapy memos, the workflow helps identify which ecDNA-amplified oncogenes warrant deeper expert review, targeted literature curation, or experimental validation, while transparently reporting uncertainty and evidence gaps.

Future work should extend this framework by integrating orthogonal molecular layers, including SNVs and indels, transcriptomic and proteomic profiles, with ecDNA structural information to refine therapeutic hypotheses. Experimental validation of prioritized candidates in ecDNA-positive cell lines and matched tumor contexts will be essential to assess functional relevance. In addition, longitudinal perturbation experiments combined with ecDNA-aware reconstruction may clarify how ecDNA dynamics influence treatment response, resistance, and disease progression over time.

Overall, we present an evidence-aware AI co-scientist workflow that connects ecDNA-amplified oncogenes to curated CIViC therapeutic evidence using PCAWG as a public pan-cancer testbed. By producing ranked, auditable candidate-therapy memos and explicitly surfacing evidence gaps, this study provides a practical and reproducible path for incorporating ecDNA biology into precision oncology research and translational hypothesis generation. ecDNA-aware interpretation offers a complementary perspective to conventional copy-number analysis by explicitly considering amplification architecture associated with intratumoral heterogeneity and rapid evolutionary adaptation. Even when the amplified oncogene is identical, ecDNA topology can imply distinct biological behavior, including dynamic copy-number changes and uneven inheritance, which may motivate alternative monitoring strategies or combination treatment hypotheses in settings where adaptive resistance is expected.

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Code availability

The code used for analysis has been deposited at

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- 9 Kim, H. *et al.* Mapping extrachromosomal DNA amplifications during cancer progression. *Nature genetics* **56**, 2447-2454 (2024).

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: yes

Justification:

- The claims regarding the integration of ecDNA structural data with CIViC evidence are supported by the analysis of the PCAWG dataset presented in the Results section.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: yes

Justification:

- We explicitly discuss the current lack of multi-omics integration and the necessity for future experimental validation in the "Limitations and Future Perspectives" section.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: N/A

Justification:

- This paper does not include new theoretical theorems or mathematical proofs; it focuses on an empirical bioinformatics workflow and data integration.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: yes

Justification

- We disclose our data sources (PCAWG) and the programmatic retrieval method for CIViC evidence, providing all necessary steps to reproduce our evidence-matching results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: yes

Justification:

- All source code for our automated querying and ranking pipeline is available on GitHub at <https://github.com/JiwonShon/pcawg-civic-ecdna-analysis>.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: yes

Justification:

- We detail the ranking criteria for drug proposals, including tumor-context match, alteration-type match, and evidence strength used in our automated engine.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: N/A

Justification:

- Our results primarily consist of descriptive prevalence summaries and qualitative evidence-matching memos rather than longitudinal statistical comparisons requiring error bars.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: yes

Justification:

- The pipeline consists of API-based data retrieval and matrix processing that can be executed on standard consumer-grade workstations; specific software requirements are listed in our repository.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: yes

Justification:

- Our research adheres to ethical standards, using de-identified, publicly available genomic data and maintaining transparency in AI-assisted curation.

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: yes

Justification:

- We discuss the positive impact of improving precision oncology for aggressive cancers while noting that AI-assisted results must be reviewed by experts to avoid clinical misinterpretation.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: N/A

Justification:

- Our work involves an open-source analytical pipeline for research purposes and does not involve high-risk generative models or restricted personal health information.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: yes

Justification:

- We properly credit the PCAWG consortium and the CIViC knowledgebase and comply with their open-access and community-curated data licenses.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: yes

Justification:

- The new analytical pipeline is documented in our open-source repository, including instructions for use and data integration.

14. **Crowdsourcing and Research with Human Subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: N/A

Justification:

- This study does not involve new human subject research or crowdsourcing; it utilizes existing de-identified data.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: N/A

Justification:

- IRB approval is not required as the study is based on publicly available, de-identified genomic datasets from the PCAWG consortium.

